THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The LCM of 30, 42 and 66 is: 1

(a) 1155 **(b)** 2310 **(c)** 6930 **(d)** 1260

2. If α, β are zeroes of $x^2 - 4x + 3$, then $\alpha^3 + \beta^3$ is: 1

(a) 40 **(b)** 28 **(c)** 64 **(d)** 100

3. The pair $kx + 2y = k - 2$ and $8x + ky = k$ has no solution when $k =$ 1

(a) -4 **(b)** 4 **(c)** 8 **(d)** 2

4. If the roots of $x^2 - 7x + 10 = 0$ are α, β, then $\alpha^2 + \beta^2 =$ 1

(a) 29 **(b)** 49 **(c)** 69 **(d)** 39

5. The least number that is divisible by every number from 1 to 6 is: 1

(a) 120 **(b)** 360 **(c)** 60 **(d)** 30

6. If α, β are zeroes of $x^2 - 5x + 6$, then $\alpha^2\beta + \alpha\beta^2$ is: 1

(a) 36 **(b)** 25 **(c)** 11 **(d)** 30

7. The pair $kx + y = k - 1$ and $4x + ky = k$ has infinitely many solutions when $k =$ 1

(a) -2 **(b)** 4 **(c)** 1 **(d)** 2

8. If the roots of $x^2 - 10x + k = 0$ differ by 2, then $k =$ 1

(a) 23 **(b)** 24 **(c)** 26 **(d)** 48

9. The largest number dividing 67, 103 and 175 leaving remainder 4 in each case is: 1

(a) 18 **(b)** 3 **(c)** 1 **(d)** 9

10. If α, β are zeroes of $x^2 - 11x + 18$, then $|\alpha - \beta|$ is: 1

(a) 11 **(b)** 14 **(c)** 7 **(d)** 8

11. If $11x + 4y = 53$ and $5x + 3y = 30$, then $x + y =$ 1

(a) 15 **(b)** 5 **(c)** 8 **(d)** 3

12. For which value of k does $x^2 + kx + 9 = 0$ have equal roots (take $k > 0$)? 1

(a) 12 **(b)** 3 **(c)** 9 **(d)** 6

13. **Assertion (A):** The HCF of 12 and 18 is 6 and their LCM is 36. 1
Reason (R): For two positive integers a and b, HCF(a, b) x LCM(a, b) = a x b.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A quadratic polynomial whose graph is a parabola cutting the x-axis at two distinct points has two real zeros. 1
Reason (R): The zeros of a polynomial $p(x)$ are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The graphs of $x + y = 4$ and $x - y = 2$ intersect at the point $(3, 1)$. 1
Reason (R): The coordinates of the point of intersection of the lines of a pair of linear equations give the solution of the pair.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Two numbers have HCF 12 and are in the ratio $3 : 5$. Find the numbers and their LCM. 2

17. If α, β are zeroes of $x^2 - 4x + 3$, form a quadratic polynomial whose zeroes are α^2 and β^2. 2

18. Find k so that the pair $kx + 2y = k - 2$, $8x + ky = k$ has no solution. 2

19. Find the values of k for which $kx^2 + 4x + 1 = 0$ has real and equal roots. 2

20. Find the smallest positive integer that leaves remainders 7 and 13 when divided by 12 and 18 respectively. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A, B and C take 18, 24 and 36 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

22. Find a quadratic polynomial whose zeroes are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, where α, β are zeroes of $x^2 - 3x + 2$. 3

23. Solve: $\frac{2}{x} + \frac{3}{y} = 2$ and $\frac{3}{x} + \frac{5}{y} = \frac{19}{6}$. 3

24. Solve for x: $\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1} = \frac{10}{3}$. 3

OR

Find the greatest number that divides 1251 and 9377 leaving remainder 1 in each case. 3

25. The zeroes of $x^2 - 10x + c$ are α and β with $\alpha - \beta = 4$. Find c. 3

OR

Find a and b for which the pair $3x + y = 7$ and $(a+b)x + (a-b)y = 14$ has infinitely many solutions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A school is arranging chairs for the annual day. There are 144 chairs in the front block and 180 chairs in the back block. The teacher wants to arrange each block in rows so that every row has the same number of chairs, and every row in both blocks has equal length. 5

(i) Express 144 and 180 as products of prime factors.

(ii) Find the greatest possible number of chairs in each row.

(iii) How many rows will there be in all?

(iv) The coordinator proposes one more block of 216 chairs with the same rule. Find the greatest row length that works for all three blocks and the total number of rows.

27. A boat goes 36 km upstream and 40 km downstream in 10 hours, and 24 km upstream and 70 km downstream in 11 hours. Find the speed of the boat in still water and the speed of the stream. 5

OR

Two taps running together fill a tank in $\frac{10}{3}$ hours. The larger tap alone takes 5 hours less than the smaller one. Find the time each takes alone. 5

END OF QUESTION PAPER — Total: 50 Marks

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